From MRI and treatment recordings to a treatment contribution on the shell grid: methods

Methods for the spatial dose model · state of the pipeline on 9 November 2026, 04:45 UTC, after the first end-to-end run of phase 2 · prepared for Soma · draft v3 for review · paths and done markers are in the companion path reference · supersedes, as a single document, the shell-grid formal definition and the treatment-data-loading definition

> **How to read this.** This is written as the methods section of the paper, so it says what we did, why we did it that way, and where it can go wrong. Most factual sentences end with a mark that says where the statement comes from: `Soma` a definition or decision made by Soma; `read` read directly from a script, a file, or output printed on the analysis machine; `derived` follows mathematically from what came before; `recalled` a published result (listed under References, not re-opened while writing); `inferred` a judgement or estimate that is not proven. Participants are named only by their de-identified M/P codes. Script names and commands live in the appendix, not in the text.

1. What we are trying to measure, and the plan

The question is simple to say and annoying to answer: **does the way the TMS electric field lands on each person's own brain relate to how their symptoms change during treatment?** `Soma` The clinical side is the PHQ-9 (24-hour version), GAD-7, the PROMIS anxiety, sleep and cognition banks, and the post-day subjective-experience items `Soma`. The physical side is what this document builds: a number per brain region, per treatment day, that says how much stimulation that region received.

Two things make that hard. First, "the same region" has to mean the same thing in every brain, without leaning on where the coil happened to sit, otherwise the predictor would be built from the outcome's neighbourhood. Second, the stimulation is not one field: it is 15 000 pulses per participant per treatment week, delivered in trains, at a frequency tied to each person's alpha rhythm, from a coil whose pose was only recorded by hand a few times per session. So the plan has two arms that meet at the end (Figure 1):

- **Anatomy arm.** MRI → head model → a fixed partition of every brain into 72 regions called **keys**, built in each person's own anatomical frame (sections 3 to 5).
- **Treatment arm.** Raw recordings → one clean file per session with every pulse time-stamped → the coil pose under every pulse train, with the gaps filled by a population model → an E-field snapshot per pose (sections 6 to 9).
- **Where they meet.** Pulses × field on the grid plane, cortex and volume, per day, under a family of weightings (sections 10 to 12), exported as per-participant tables and set against the clinical, comfort and run-sheet covariates (sections 12 and 13).

The chain has now run end to end on the twelve phase-2 participants, and runs unattended for any participant whose treatment finishes: each stage waits for the previous stage's done marker `read`. The phase-1 participants are going through the same chain as a second set `Soma`. The longer aim is to accumulate enough participant-days to train a classifier on the contribution and its covariates `Soma`.

The order of the treatment arm is fixed: (1) raw recordings to preprocessing; (2) clinical database values, trigger timeline and gap-fill decisions; (3) one field snapshot per known, inferred or stochastic pose; (4) contribution per key; (5) the annotated recordings go back to network storage `Soma`.

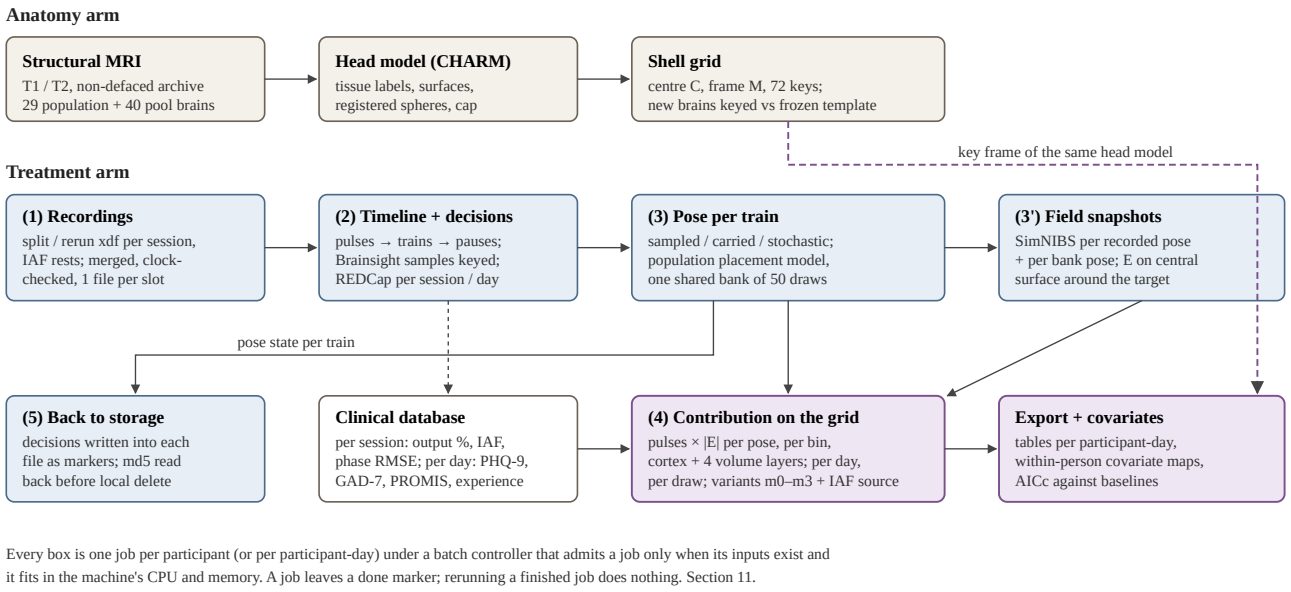


Figure 1. The pipeline. Brown: anatomy. Blue: treatment recordings. Purple: where the two meet. The dashed purple line is the only place anatomy enters the dose: the key frame (centre, rotation, key assignment) of the head model on which the field was simulated.

2. Participants and data

Head models. Two sets of brains are used read:

- **The population** is the 29 participants with head models built before this work (M017 to M034, P101 to P117). The key template is averaged over them, and every validation in section 5 is on them.
- **The archive pool** is 40 further brains taken from the non-defaced study archive and re-identified with random pool numbers, so that no working file links them back to study codes Soma. The defaced copies of the same scans are not used, because defacing cuts the scalp that the head model needs read. One person in the pool has a repeat scan, which is kept under the same pool number with a session tag. These 42 head-model jobs are keyed against the frozen population (section 4.7), so adding them never moves the population's keys.

Treatment recordings. Twelve phase-2 participants have treatment recordings read. Each had 5 treatment days with 6 sessions a day, so 30 sessions, and each session delivered **3000 pulses as 75 trains of 40** read. We had been told 80 trains and 3000 pulses, which does not multiply out; the recordings settled it read. Pulses are delivered by a closed-loop system that targets a phase of the ongoing EEG at the participant's **individual alpha frequency** (IAF), about 9 Hz in the pilot, so pulses within a train are about 110 ms apart and a train lasts roughly 4.4 s read. A train cycle is about 19 s, which leaves about 14 s between trains read. Before every session there is a 5-minute resting recording used to estimate IAF, plus a morning baseline and sometimes a rest after session 6 read.

The pilot participant is **P116**: full five days, Brainsight project available, and every step was checked by hand on it first Soma. P115 is the worked example for the anatomy (its head model existed first) and is also a treatment participant, processed from a separate copy of its recordings read.

Clinical and run-sheet data. Everything comes from the study's REDCap export (PHASE II, 7 November) converted by Soma's existing converter, so nothing here re-pulls or re-derives clinical data Soma. Per session it provides the stimulator output (% of maximum), the IAF used, and the phase RMSE of the closed-loop predictor. Per treatment day it provides the pre-day self-report (PHQ-9 24 h, GAD-7, PROMIS T-scores), post-day items, and, for each outcome, its next-day value and change read. The pre-day events map to days by suffix: self-report events '', c, f, h, j and safety events b, d, e, g, i are days 1 to 5 read. For orientation, P116's PHQ-9 (24 h) over days 1 to 5 was 18, 17, 17, 15, 15 read.

Phase 1. Twenty phase-1 participants (M016, M018, M020 to M037) had the same protocol as far as this pipeline is concerned Soma. Nineteen have treatment recordings in the de-identified study archive; one has none read. A structural comparison against phase 2 found the same six streams with the same channel layouts and labels, 40 pulses per train and 75 trains per complete session. The differences are that the phase-1 rest recordings have no marker stream and last 300 s instead of 600 s, the file headers are on US Eastern time, and one participant has a seventh session slot read. Fourteen are in the chain now; five need their neuronavigation projects extracted and their field caches built first read. No phase-1 result is in this version.

3. Head models

Each structural MRI (T1, with T2 where present) goes through **CHARM**, the head-modelling tool of SimNIBS 4.6 read recalled (Puonti et al. 2020). From one participant's output we use read:

- a **tissue label volume**, one label per voxel (white matter, grey matter, cerebrospinal fluid, and so on);
- per hemisphere, the **white**, **pial** and **central** surfaces as triangle meshes; the central surface runs through the middle of the cortical ribbon;
- per hemisphere, the **registered sphere**: the same mesh inflated to a sphere and warped so its folding pattern lines up with the FreeSurfer average brain recalled (Fischl et al. 1999);
- **EEG electrode positions** for a 64-channel cap, which are a template cap warped onto the head through CHARM's registration to standard space, not measured on the person read.

A mesh is a vertex set $\mathcal{V}$ with positions $\vec{x}_v$ and a list of triangles. The area of vertex v, a_v, is a third of the area of the triangles touching it, so $\sum_v a_v$ is exactly the mesh area derived. The **medial wall**, the cut that closes each hemisphere's sheet through the deep midline, is not cortex and is excluded from every cortical measure read. Each vertex gets a Desikan–Killiany atlas label by nearest neighbour on the registered sphere read recalled (Desikan et al. 2006). Labels describe and check the keys; they never define them.

Batch numbers. Under the batch controller each archive head model was given 2 cores and used both fully, peaked at 9.5 GB of memory on average (10.9 GB at most), and took a median of 76 minutes read. CHARM is therefore CPU-bound here, and future batches should declare about 11 GB and 3 to 4 cores per brain inferred.

4. The shell grid

The grid's job is to give "the same place" a meaning across brains that does not depend on the coil. It does that by **direction from a centre**: a key is a cone with its tip at the centre of the brain, pointing in the same anatomical direction in everyone. The folding inside the cone is allowed to differ between people, and is measured rather than forced to agree Soma.

A useful way to hold it in your head: think of the brain as seen from its own middle. Every point is "which way" plus "how far". The keys cut up "which way"; shells cut up "how far".

4.1 The centre

Take both pial surfaces, each a closed, outward-oriented triangle mesh with triangles $t = (\vec{a}_t, \vec{b}_t, \vec{c}_t)$, and any reference point $\vec{o}$. Each triangle with $\vec{o}$ forms a tetrahedron of signed volume $V_t = \frac{1}{6}\det[\vec{a}_t - \vec{o},\ \vec{b}_t - \vec{o},\ \vec{c}_t - \vec{o}]$, and

$$V = \sum_t V_t, \qquad \vec{C} = \frac{1}{V}\sum_t V_t\,\frac{\vec{o} + \vec{a}_t + \vec{b}_t + \vec{c}_t}{4}.$$

By the divergence theorem V is the volume enclosed by the pial surfaces and $\vec{C}$ is its centroid, whatever $\vec{o}$ is derived. The enclosed solid is the cerebrum; the cerebellum and brainstem are outside it derived. We started with the midpoint of

the anterior and posterior commissures, mapped in from standard space, but in four participants (M030, P103, P108, P110) that point landed more than 33 mm below the cortex centroid read. Moving the centre by δ turns a boundary at distance r by about δ/r radians, so that would have rotated keys by more than 20° derived. Soma asked for a centre from each person's own anatomy instead Soma.

4.2 Direction, distance and the anatomical frame

For $\vec{x} \neq \vec{C}$,

$$r(\vec{x}) = \|\vec{x} - \vec{C}\|, \qquad \pi(\vec{x}) = \frac{\vec{x} - \vec{C}}{r(\vec{x})} \in \mathbb{S}^2, \qquad \vec{x} = \vec{C} + r(\vec{x})\,\pi(\vec{x}).$$

Directions must be measured against axes that belong to the brain, not to the scanner or the cap. Each participant gets three axes:

Outer radius. For a direction $\hat{u}$, $r_{\text{out}}(\hat{u})$ is the largest distance from $\vec{C}$ among pial vertices within 3° of $\hat{u}$, sampled on 4000 near-uniform (Fibonacci) directions and smoothed to $\tilde{r}$ by least squares on polynomials of degree ≤ 4 in the components of $\hat{u}$, which on the sphere are the spherical harmonics of degree ≤ 4 read recalled.

Pole. The outer surface $\vec{C} + \tilde{r}(\hat{u})\hat{u}$ has normal along $\hat{u} - \nabla_S \ln \tilde{r}$, so the ray from the centre is perpendicular to it exactly at critical points of $\tilde{r}$ derived. Soma defined the pole as that point near the top of the head Soma:

$$\hat{p} = \operatorname*{arg\,min}_{\angle(\hat{u},\hat{z}) \leq 45^\circ} \tilde{r}(\hat{u}).$$

It is a minimum, not a maximum: seen from its centre a brain is longest front to back and shortest to the top, like an ellipsoid with $a > b > c$ seen from its centre derived. Both Hessian eigenvalues of $\tilde{r}$ at the pole are positive in all 29 participants (69 to 127 mm/rad²) read. An earlier version that took any critical point landed on lateral saddles in 10 people, which is why the minimum is required read.

Left–right and the rotation. With $\vec{c}_L, \vec{c}_R$ the area-weighted centroids of the two central surfaces and $\vec{\ell} = \vec{c}_R - \vec{c}_L$,

$$\hat{x} = \frac{\vec{\ell} - (\vec{\ell}\cdot\hat{p})\hat{p}}{\|\vec{\ell} - (\vec{\ell}\cdot\hat{p})\hat{p}\|}, \qquad \hat{y} = \hat{p} \times \hat{x}, \qquad M_p = [\hat{x}\ \ \hat{y}\ \ \hat{p}].$$

$\hat{x}$ points right, $\hat{y}$ anterior derived. The medial-wall plane was tried first and was useless: its normal sits 84° to 90° from left–right in everyone, because the medial wall is a cut, not a flat midline read.

Coordinates. With $(u_x, u_y, u_z) = M_p^\top \hat{u}$,

$$\varphi = \operatorname{atan2}(u_y, u_x) \in [0^\circ, 360^\circ), \qquad \zeta = u_z = \cos\psi \in [-1, 1].$$

So $\varphi = 0^\circ$ is right, 90° anterior, 180° left, and ψ is the angle from the pole Soma. Soma chose $\zeta = \cos\psi$ rather than ψ because the solid-angle element is then $d\Omega = d\varphi\, d\zeta$: equal rectangles in (φ, ζ) are equal solid angles (Archimedes' hat-box theorem) recalled.

4.3 Seeds and keys

The outer shell S_1 carries the EEG electrode layout; each electrode marks the centre of one key Soma. For participant p and electrode k at $\vec{e}_k^{\,p}$ the direction in the participant's own frame is $\hat{u}_k^{\,p} = M_p^\top(\vec{e}_k^{\,p} - \vec{C}_p)/\|\cdot\|$, and one **template** serves everyone:

$$\bar{u}_k = \frac{\sum_p \hat{u}_k^{\,p}}{\|\sum_p \hat{u}_k^{\,p}\|}, \qquad \text{seed of } k \text{ in } p = M_p\,\bar{u}_k.$$

63 electrode names occur in all 29 caps read. The cap only enters through the average, so independent placement errors δ shrink to about $\delta/\sqrt{29} \approx \delta/5.4$ derived. The pilot grid was refined near the stimulation target; Soma removed that, because it built a prefrontal bias into the predictor itself Soma.

Assignment. Each point goes to the nearest seed by direction:

$$\kappa(\vec{x}) = \arg\max_g \ \pi(\vec{x}) \cdot M_p \bar{u}_g = \arg\min_g \ \|\pi(\vec{x}) - M_p \bar{u}_g\|,$$

the two forms agreeing because $\|\vec{a} - \vec{b}\|^2 = 2 - 2\,\vec{a}\cdot\vec{b}$ for unit vectors derived. On the sphere this is the spherical Voronoi diagram of the seeds recalled (Du et al. 1999). In space, each key

$$K_g = \{\vec{C} + \vec{y} : \vec{y} \cdot (M_p \bar{u}_g - M_p \bar{u}_h) \geq 0 \ \ \forall h \neq g\}$$

is an intersection of half-spaces through $\vec{C}$: a convex polyhedral cone derived. The keys partition the brain up to boundary planes of zero volume, and they do so **without** needing any surface to be star-shaped: a ray that crosses folded cortex five times stays in one key, because the key depends only on direction derived. That property is the reason this design replaced the earlier straight-ray and peak/pit designs, where rays from the centre reached the right piece of cortex first for only 36 to 40% of vertices read.

The volume of key g between radii $r_1 < r_2$ is $\Omega_g(r_2^3 - r_1^3)/3$, with Ω_g the solid angle of its cell derived.

Midline split. A cell whose template seed is within 8° of the midsagittal plane ($|\bar{u}_{k,x}| < \sin 8^\circ$) would hold both hemispheres, so its vertices go to a left and a right key by hemisphere read. Nine seeds are split, which turns **63 seeds into 72 keys** read. From here on g is one of the 72.

4.4 Shells and the three pieces of a key

For key g, $\rho_2(g)$ is the largest distance from $\vec{C}$ among its pial vertices and $\rho_3(g)$ the smallest among its white-surface vertices, medial wall excluded read. These define Soma's shells $S_i = \{\vec{C} + \rho_i(\kappa(\hat{u}))\hat{u}\}$, $i = 2, 3$: the outer envelope of the cortex and the envelope beneath the deepest sulci Soma. With B the brain tissue in the label volume,

$$V_g^{\text{in}} = \{\vec{x} \in K_g \cap B : r < \rho_3(g)\}, \quad V_g^{\text{ctx}} = \{\rho_3(g) \leq r \leq \rho_2(g)\}, \quad V_g^{\text{out}} = \{r > \rho_2(g)\}.$$

These always partition the key derived. The meaning "all cortex is in the middle piece" holds by construction, since distance from a point is convex along a straight edge, so no pial triangle reaches past its farthest corner derived. The shells are terraced, constant within a key, not smooth derived. In keys through the insula or under the Sylvian fissure the band $\rho_2 - \rho_3$ is several centimetres thick and holds white matter; Soma's decision is that overhanging cortex is flagged, not corrected Soma.

4.5 What is measured in each key

With $\mathcal{V}_g$ the key's central-surface vertices (medial wall removed):

- cortical area and mean distance: $A_g = \sum_{v \in \mathcal{V}_g} a_v$, $\bar{r}_g = A_g^{-1} \sum_v a_v\, r(\vec{x}_v)$;
- atlas label fractions $w_{g\ell} = A_g^{-1} \sum_{v:\lambda(v)=\ell} a_v$ and the majority label $\ell_g^* = \arg\max_\ell w_{g\ell}$ Soma;
- folding index $F_g = A_g / (\Omega_g^{\text{occ}}\, \bar{r}_g^{\,2})$, the cortical area over the area of the sphere patch it projects onto, counted in equal-area bins of 2° × 0.022 read. It is scale-free, and it also grows where rays graze the cortex, so it mixes folding with obliquity derived inferred;
- depth below a smoothed outer envelope $\varepsilon(\hat{u})$, along the ray: $d(v) = \varepsilon(\pi(\vec{x}_v)) - r(\vec{x}_v)$; crowns and fundi are the shallowest and deepest 10% of each key's vertices read.

Volume descriptors on the angle plane. For the per-brain figures (section 4.6) two fields are drawn over the plane (φ, ζ):

$$I(\hat{u}) = \int_0^{r_{\text{out}}(\hat{u})} \rho\big(\vec{C} + r\hat{u}\big)\, r^2\, dr \quad [\text{ml/sr}], \qquad \|\nabla_S I\|,\ \ \|\nabla_S r_{\text{out}}\|,$$

where ρ is the brain-tissue indicator of the label volume inferred (the figure says “tissue integrated over each cone”; whether grey and white matter are weighted differently has not been re-read). I is the tissue volume per unit solid angle along a direction, because the volume element in spherical coordinates is $r^2\, dr\, d\Omega$ derived. Its gradient is large where the amount of tissue changes fast with direction, which is at sulcal walls and at the edges of the temporal lobes, so it shows the folding that the keys deliberately do not follow inferred.

4.6 One brain, start to finish

Figure 2 follows P115. Its centre lies at $(-0.06, -14.04, 6.52)$ mm in scanner coordinates; its pial surfaces enclose 1044 ml, while its white plus grey matter voxels total 1220 ml, because the label volume also counts the cerebellum and deep grey matter read. Its pole is 1.95° from the scanner vertical, and the outer radius there is 50.7 mm read. All 72 keys contain cortex; key areas run from 217 to 7023 mm², median 2655 mm² read. Key 41 on the left, over the former stimulation target, has 2488 mm² of cortex, 93% of it rostral middle frontal, and a folding index of $2488/(0.136 \times 66.5^2) = 4.14$, which you can check by hand read derived.

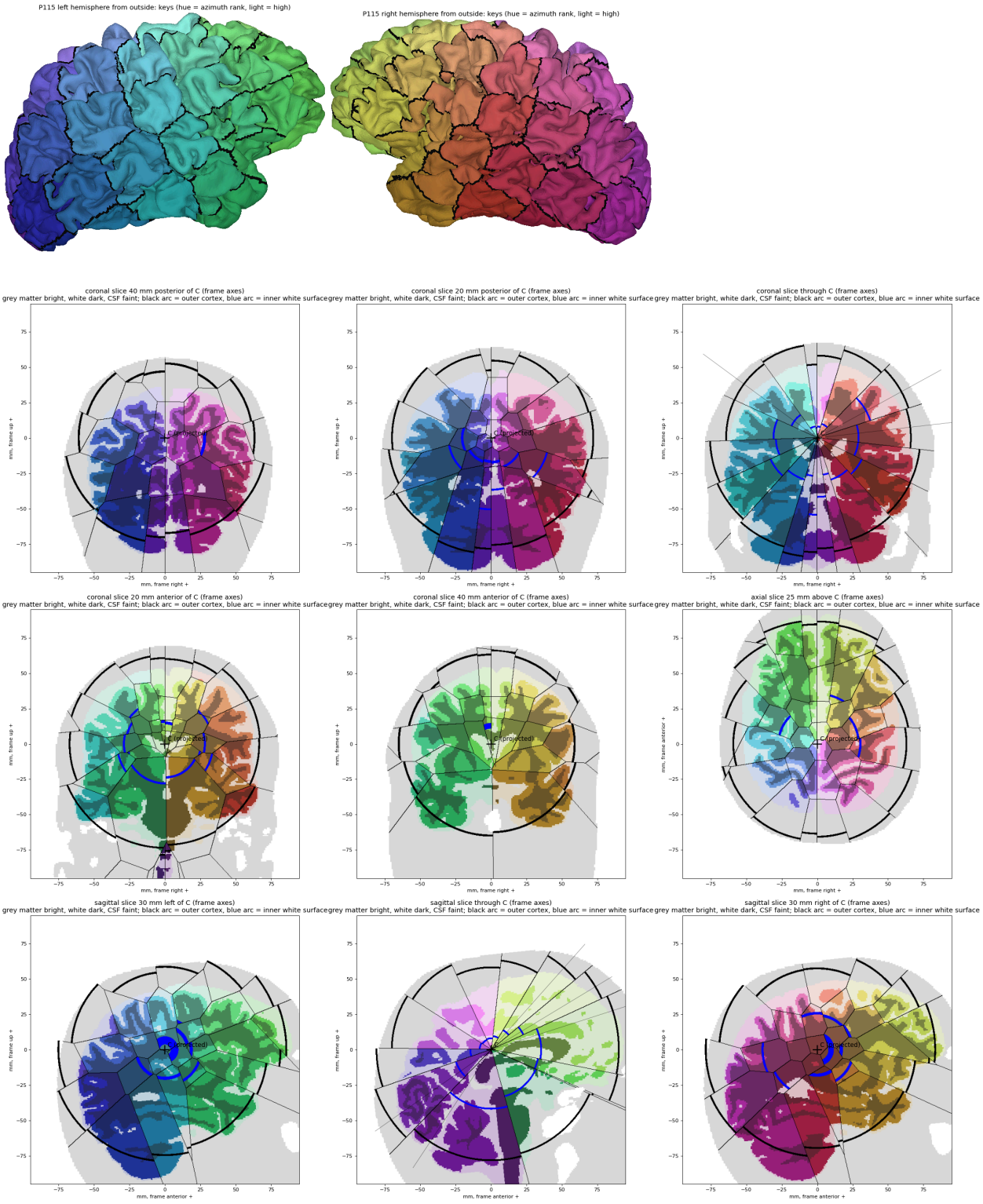


Figure 2. P115 keyed by the shell grid, in volume. *Top:* each hemisphere's pial surface from outside, coloured by key (hue = azimuth rank, lightness = height), with black key borders. *Below:* coronal slices 40 and 20 mm posterior, through, and 20 and 40 mm anterior of the centre, one axial slice 25 mm above it, and sagittal slices 30 mm left, through and 30 mm right of it, all in the participant's own frame. Grey matter is bright, white matter dark, cerebrospinal fluid faint. Black arcs are the outer cortex shell S_2, blue arcs the inner white shell S_3, and the thin lines are key boundaries converging on $\vec{C}$. Where a slice passes through $\vec{C}$ the boundaries meet at one point, which is the cone structure of section 4.3.

4.7 New brains: keyed against the frozen population

A head model that arrives after the population run is keyed **against the frozen template**: the 63 template directions and the 72-key list are read, and only that brain's own centre, frame and assignment are computed `read`. The population's keys never change when a brain is added `derived`. Each added brain writes a check record: keys present out of 72, the

share of keys whose majority atlas label matches the population's, and whether the pole is a true minimum inside its search cap read. That label share is the first test of the grid on brains that did not help build it. All 42 archive head models have been keyed this way read.

5. Does the grid hold up across people?

This section is about honesty more than results, because a partition that is identical by construction will always look beautiful.

What is true by construction and carries no evidence. In the frame, both the seeds and the directions are rotated by the same M_p, so the partition drawn on the sphere is the same template Voronoi diagram in everyone. "Every participant has the same partition" is therefore not a finding, and that panel was removed derived. Having all 72 keys present in all 29 people is close to automatic too derived.

What is measured. Things not fixed by the grid, compared across people read:

Run	Frame and grid	Label agreement (median over keys)	Same registered location (median over pixels)	Key-area CV	Relative-area CV
earlier	each person's own CHARM cap	0.72	n/a	0.26	0.25
final	**anatomical frame, uniform template**	**0.97**	**0.83 lateral / 0.79 top**	**0.12**	**0.09**

Label agreement is, per key, the share of participants whose majority atlas label equals the most common one. *Same registered location* takes one pixel on the unfolded registered sphere, which is the same anatomical place in everyone, and asks what share of participants have the same key there; at key borders it falls to 0.52 read. Both measures inherit the surface registration's own error, so they mean "agrees with what the registration calls the same place", not ground truth derived.

Effective sample size. Many vertices are not many tests. A 3° frame error moves all 72 keys of a person at once; the free quantities per person are the centre (3), the rotation (3) and a smooth warp, so the number of independent comparisons is somewhere between about 6×29 and 72×29 derived.

5.1 Rotation null

If all brains are roughly the same shape, the agreement might come from any sensible orientation. To test that, each participant's template directions are turned by an angle θ about an axis drawn uniformly at random, **a different axis per person** (a rotation shared by everyone would leave agreement unchanged), the brain is re-keyed, and both agreements are recomputed with the population code derived. $\theta \in \{1, 2, 4, 8, 16, 32\}^\circ$ with 20 draws each, plus a uniformly random orientation per person, which gives the chance floor read. The floor is not zero because among 29 unrelated answers the most common one always collects a few derived.

Frame error	0°	1°	2°	4°	8°	16°	32°	random
Label agreement	0.97	0.97	0.94	0.90	0.79	0.57	0.31	0.13
Same location, lateral	0.83	0.81	0.79	0.76	0.64	0.43	0.22	0.10
Same location, top	0.79	0.79	0.79	0.75	0.63	0.43	0.22	0.10

read. Agreement is flat to 2°, halves (between real value and floor) at about 17° for labels and 15° for the same-location measure, and no random draw comes near the real grid derived. Each person's own frame error, estimated as their best-fitting global rotation of the atlas regions, has median 2.0° and maximum 4.1° (P108), which costs at most 0.07 in label agreement read derived. The old cap-based grid's 0.72 corresponds to a frame error of about 10°, which matches the front-to-back scatter of the CHARM cap vertex about the anatomical pole (SD 7.8°) inferred. So the jump from 0.72 to 0.97 is the frame, not luck.

What the test cannot show: it perturbs only the orientation, not the centre position or the template; and consistency between people cannot tell whether key 41 sits over a real electrode 41 derived.

Validation against other methods, 40-brain pool. The geometry the grid measures is reproducible (reliability of the outer shell radius about 0.98, of crown position about 0.82), but majority-label agreement cannot tell the same person from two different people read. So the keys are a shared coordinate system, not an individual fingerprint, and no claim should rest on a person's key identity alone derived.

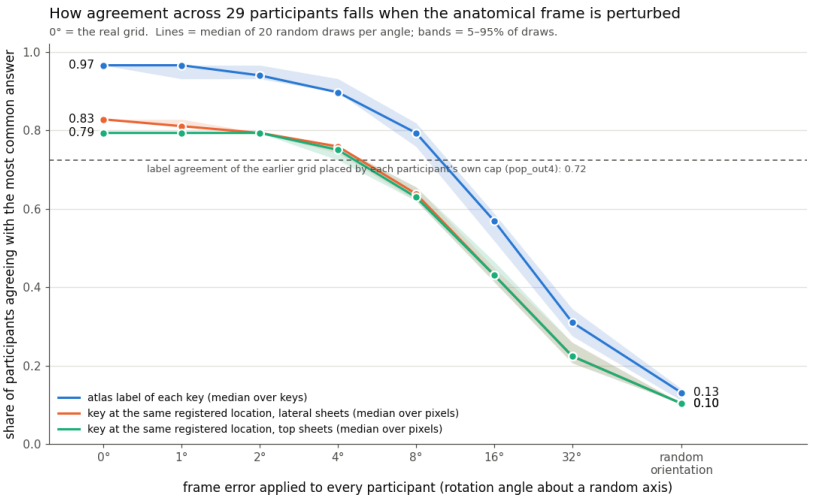


Figure 3. Agreement against a simulated frame error. Each participant's frame is rotated about its own random axis. Blue: label agreement (median over keys). Orange and green: agreement at the same registered location on the lateral and top sheets (median over pixels). Lines are medians over 20 draws, bands the 5th to 95th percentiles; "random orientation" is the chance floor. Dashed: the earlier cap-based grid (0.72).

5.2 Where the variability really is: displacement and smearing

Keys are not meant to follow folds. Soma models how a brain region differs between people within the keys by three stochastic terms: a **displacement in azimuth**, a **displacement in height**, and a **smearing** σ_{Sm} of the region's area outward along the keys Soma. For each atlas region and person we take the area-weighted mean direction in the frame and compare it with the population mean, before and after removing each person's best global rotation read:

- total displacement 2.31° east, 2.38° north; local (after the rotation) 2.12° and 2.02°; east and north correlate at 0.02, so the two angle terms are independent, as Soma assumed read derived;
- smearing $\sigma_{\mathrm{Sm}} = 0.023$ globally and 0.028 locally, with area growing with distance from the centre as $r^{2.73}$ read.

A 2° to 3° displacement against keys about 29° across is a ratio of about 10, which is why agreement is high inside keys and drops only near borders derived. The folds themselves are individual: at the same registered location, depth varies with a median SD of 3.0 mm, and depth and curvature deviations correlate only about −0.4, so they carry different information read.

Allometry, briefly. Cortical area scales with brain volume with exponent 0.97 (95% interval 0.78 to 1.17), which excludes the geometric-similarity value 2/3 but cannot separate published adult values of 0.8 to 0.9 read recalled (Im et al. 2008; Toro et al. 2008). No key's area slope differs from 1 after Benjamini–Hochberg correction; 2 keys' folding index depends on size read.

The CHARM cap. Against caps digitised during sessions, the CHARM cap sits 3.4° and 4.3 mm from each person's mean real cap, the same as a single real session (3.6°, 4.3 mm), so its placement is not systematically off; its shape is (per-electrode residual 8.5 mm vs 2.9 mm) read. Soma took CHARM's ideal placement as the reference for both the grid and the EEG Soma.

6. Treatment recordings: one clean file per session

The acquisition writes one file per run, and a session that was stopped and restarted is split into several runs; one session of P116 had six read. Before any analysis we rebuild the recordings so that each **slot** (one session or one rest) is one file, on one clock, with every pulse listed. This step is structural only: EEG sample values are copied unchanged, byte for byte in dtype read.

6.1 Indices

For one participant read:

- **day** $d \in \{1..5\}$: the rank of the local calendar date (America/Chicago) among the dates that have treatment runs. Local, not UTC, so a late session stays on its own day;
- **slot** $j \in \{1..6\}$: the session number in the run's file name;
- **session** $k = 6(d-1) + j \in \{1..30\}$;
- **file index** $i = 13(d-1) + m$, with $m = 2j - 1$ for the rest before session j, $m = 2j$ for session j, and $m = 13$ for the rest after session 6. Missing rests leave holes in i by design;
- **train** r, numbered in time within a session across continuous joins; **pulse** $q \in \{1..40\}$ within a train.

6.2 Merging runs and the clock

The input set for slot (d, j) is every run whose local date ranks d and whose name carries slot j, ordered by first time stamp read. Time stamps are the recorder's synchronised clock (LSL), with no dejittering read.

Joins. Between consecutive runs a, b, let g be the first sample of b minus the last sample of a on the fastest stream. The join is **continuous** if $g < 1$ s (the recording was split and nothing happened) and a **restart** otherwise; $g < 0$ is flagged as overlap read. On P116 day 1 every join was a restart, with gaps of 14 to 226 s read. File-name times are not used for timing: they lag the recording by up to about 5 s and once made every restart look continuous read.

Clock check. For each run f, $o_f = u_f - \tau_f$, the file-name Unix time minus its first synchronised stamp. o_f varies by a few seconds (recorder start latency). A run with $|o_f - o_{f_1}| > 60$ s is a clock reset; its stamps are shifted and a marker records the shift read. Unix time is then

$$t_{\text{unix}} = t_{\text{LSL}} + \text{median}_f\ o_f,$$

so **relative times are exact and absolute times good to a few seconds** read.

Markers. A new marker stream records run starts and ends, each join with its type and gap, clock shifts, and the session's first, 3000th and last pulse read. Every pulse becomes a row with its run, train, position in the train, LSL and Unix time read.

Pulse counts. On P116 day 1, sessions 2 to 6 have exactly 3000 pulses (75 × 40); session 1 has 2998, made of aborted partial trains at the start (25 and 13 pulses) followed by 74 full trains read.

Publishing. Per day: convert and verify locally (re-read, hash every stream), annotate with the pose decisions (section 8), copy to network storage, **read every copy back and require its md5 to match**, write the day summary last (its presence means the day is published), and only then delete the local copy Soma read. The raw source is never written read. Publication is the **last** step for each participant, after the contribution, so a slow network never holds up the

analysis Soma. Phase-1 raw recordings are pulled from the archive to a staging disk, at most three participants at a time, converted there with the Eastern time zone, published, and the staged copy removed read.

7. Clinical database join

For session k the run sheet gives the stimulator output O_k (% of maximum), the IAF f_k (Hz) and the predictor's phase RMSE ϵ_k read. For day d and outcome o it gives $Y_{d,o}$, the next-day value, and the change

$$\Delta Y_{d,o} = Y_{d+1,o} - Y_{d,o}.$$

Pre-day measures are taken before that day's first session, so $\Delta Y_{d,o}$ is the change across treatment day d inferred. These values are written into each published file's header and into a per-day table, so the recordings carry their own clinical context read. Each value is joined by (k, d), and a mismatch between the REDCap date and the recording date is reported, never silently shifted read.

8. Coil pose under every pulse train

This is the part where we have the least data and have to be most careful about what is measured and what is assumed.

8.1 What Brainsight actually stored

The neuronavigation project file holds a table of hand-taken **samples**: 48 rows for P116's whole treatment, 1 to 4 per session. There is no per-pulse coil tracking; the trigger input was configured but not used to record poses read. Each sample is a 4×4 coil pose $(\mathbf{R}, \vec{x})$ and the target it was taken against, $(\mathbf{R}_{\text{ref}}, \vec{x}_{\text{ref}})$ read. Soma's existing extraction keeps 47 of them, merges duplicates and fixes slot names read.

Clock. Brainsight writes local time with no zone. We choose the whole-hour offset $h \in [-12, 12]$ that puts the most samples inside their own session window $[t_{\text{first}} - 30\ \text{min},\ t_{\text{last}} + 5\ \text{min}]$:

$$t_s = t_{\text{BS}} + 3600\,h + \delta.$$

$h = -5$ wins, with 39 of 47 samples inside read; assigning each sample to a session by time window then places 47 of 47 read. The residual δ is not identified: one estimate said −480 s, another −190 s; we use $\delta = 0$ read. Eight day-2 samples sat one session later than their names said, and session 7 has no sample at all read.

Deviation from target. Rotations are first projected onto SO(3) by SVD. In the target frame,

$$\vec{o} = \mathbf{R}_{\text{ref}}^{\top}(\vec{x} - \vec{x}_{\text{ref}})\ [\text{mm}], \qquad \vec{w} = \log\!\left(\mathbf{R}_{\text{ref}}^{\top}\mathbf{R}\right)^{\vee}\ [\text{deg}], \qquad \vec{d} = (\vec{o}, \vec{w}) \in \mathbb{R}^6,$$

where $\log(\cdot)^{\vee}$ is the rotation vector (axis × angle) read recalled. For P116, the coil centre sat 0 to 7.6 mm from target and the coil normal 0 to 10° from the target normal read. One session-1 sample has exactly zero deviation because the operator used it as the target; it is real but not a real deviation read.

8.2 Pauses

From the pulse table, train r of session k has first and last pulse $t_0(k,r)$, $t_1(k,r)$ and gap $g_r = t_0(k,r) - t_1(k,r-1)$. Train r **starts a pause** if read Soma

$$\big(\text{run}(r) \neq \text{run}(r-1)\ \wedge\ g_r \geq 1\ \text{s}\big) \quad \text{or} \quad g_r > 60\ \text{s}.$$

The first case is a restart between merged runs, the second a long gap within a run. Call $t_r = t_0(k,r)$ the resume time. **Gaps in the population** read. Over all twelve participants on LSL timing there are 26 646 gaps, about 2220 per participant, which is 30 sessions × 74 (75 trains less one). The median gap is 11.0 s, the middle half lies between 8.3

and 14.7 s, the 90th and 95th percentiles are 19.8 and 25.0 s, and the 99th is 57.9 s. The mean is 12.2 s, or 11.1 s without the gaps over 60 s. So the 60 s rule sits at the 99th percentile and marks 257 gaps as pauses, with a median of 98 s and a tenth of them longer than 231 s. Participant medians range from 8.6 to 13.8 s. The spread of normal gaps is the closed-loop system waiting for the right phase inferred. At least a tenth of all "gaps" are 0.1 s, one inter-pulse interval: either trains fired back to back or one train split in two by the train numbering. Which of the two has not been checked, and these are not gaps in any physical sense read inferred.

The sampling problem this creates. A participant has about 2220 gaps and about 28 Brainsight samples, of which about 21 gaps per participant exceed 60 s read. The samples cannot be matched to gaps; they are sparse measurements of a pose that is assumed constant between pauses. The pose rule below is one answer; section 12 treats the uncertainty it leaves as a variant rather than as a fact to be resolved. On P116, the four samples that fell inside gaps all fell in short, normal gaps, so a recorded sample does not imply a long pause, and the long gaps (70 to 220 s) were mostly restarts read.

8.3 The pose rule, in three layers

Every train gets a pose in three layers, and every output row keeps all three, so any later analysis can ignore the rule or the fill read:

1. **Raw.** The latest sample with $t_s \le t_0(r)$, from any session. No rule, no choice (Figure 4).
2. **Rule** (Soma's) Soma read. With $w = 5$ min, a tolerance of 30 s after resume, and "best" meaning smallest $\|\vec{o}\|$ (orientation is free for comfort, so it does not enter the choice):

Where	Condition	State
session start	best own sample in $[t_{\text{first}} - 10\ \text{min},\ t_{\text{first}} + 30\ \text{s}]$	sampled
pause	best own unused sample, or the pose in force, sampled in $[t_r - w,\ t_r + 30\ \text{s}]$	sampled
between pauses	an unused own sample in $(t_0(r-1), t_0(r)]$	sampled mid-run, from train r
between pauses	otherwise	carried
session start or pause	none of the above	**stochastic**

3. **Fill.** Every stochastic segment takes its pose from the population placement model (section 8.4).

Why the rule looks like this: a sample is the operator's check after adjusting the coil, it belongs to its own session, and an unsampled pause means the coil may have moved Soma. The 10-minute window at session start is wider because one session's setup sample was taken 5.8 minutes before the first pulse read.

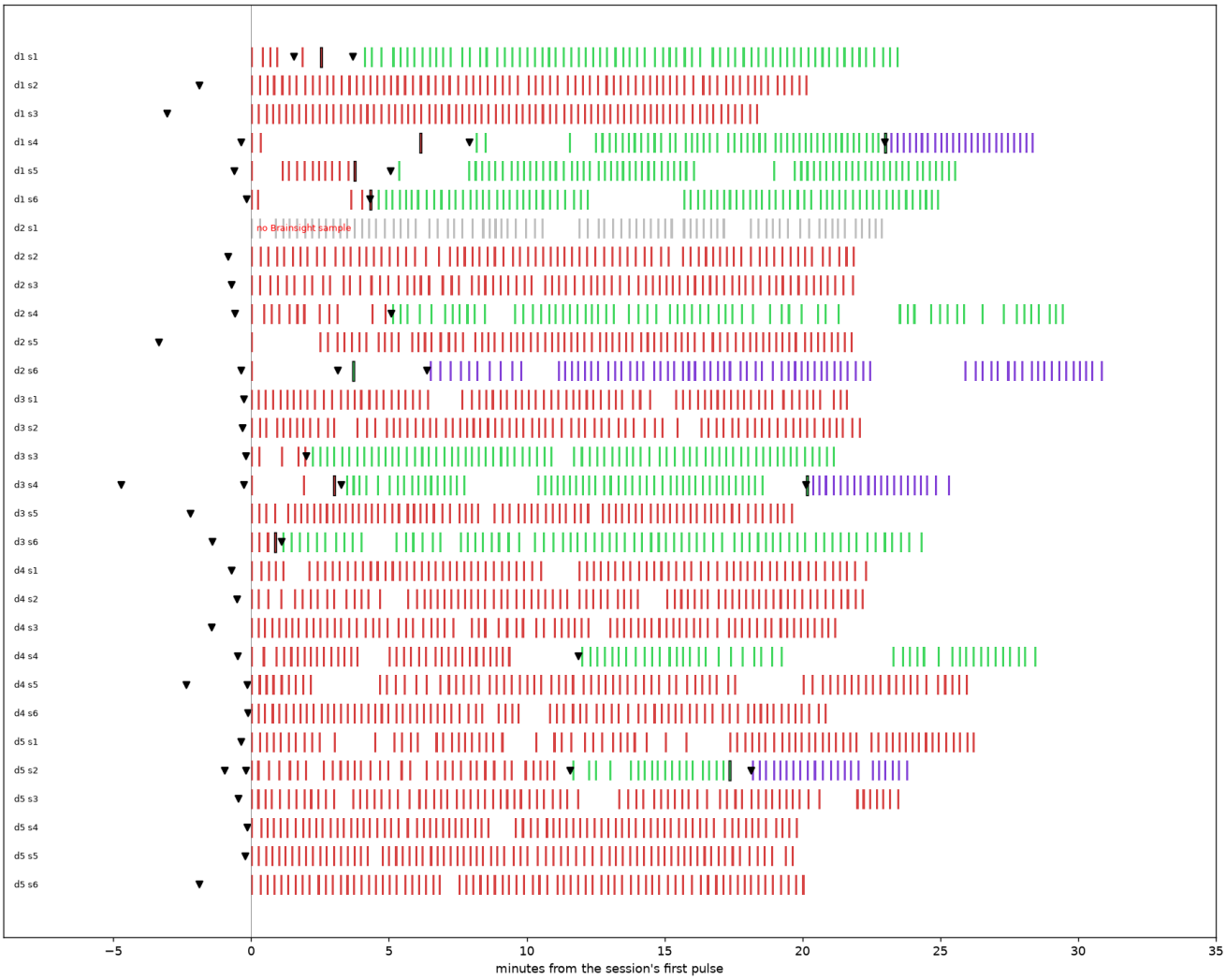


Figure 4. Layer 1 for P116: each pulse train coloured by the Brainsight sample in force when it was delivered (outlined = the pose changed mid-train; ▼ = sample time, on the Brainsight clock at UTC−5). One row per session, minutes from the session's first pulse. Day 2 session 1 has no sample.

8.4 Population placement model

The first version of the fill was per participant: a Gaussian over that person's own recent deviations, weighted by recency. Soma replaced it with **one placement model across all participants**, used wherever the rule leaves a gap, with **one shared bank of random draws**, so that differences in dose between people come from how they were placed and from their anatomy, not from different random numbers Soma.

Model. For sample i of participant p with deviation $\vec{d}_{pi} \in \mathbb{R}^6$,

$$\vec{d}_{pi} = \vec{\mu} + \vec{b}_p + \vec{e}_{pi}, \qquad \vec{b}_p \sim \mathcal{N}(\vec{0}, \mathbf{T}), \qquad \vec{e}_{pi} \sim t_\nu(\vec{0}, \mathbf{S}),$$

where $\vec{\mu}$ is the average placement, $\vec{b}_p$ the participant's own systematic offset and $\vec{e}_{pi}$ the session-to-session scatter read. The scatter is Student-t because the data are heavy-tailed: on 770 samples from 28 participants the kurtosis of the offsets reached 136, driven by a few large misplacements (up to 77 mm) and by real comfort rotations of 24° to 48° about the coil normal with under 10 mm of offset read. Those rotations are deliberate operator choices, so they are kept and down-weighted, not cut Soma.

Fit read. ν is chosen by maximum t likelihood over $\{3, 4, 6, 10, 30, \infty\}$, with $\nu \geq 3$ so the variance is finite. In the fit each sample carries weight $u_{pi} = (\nu + 6)/(\nu + m_{pi}^2)$, with m_{pi}^2 its Mahalanobis distance under $\mathbf{S}$, the usual

EM weight of a multivariate t recalled. With $\bar{\vec{d}}_p$ the weighted mean of participant p's n_p samples,

$$\hat{\vec{\mu}} = \mathrm{mean}_p\, \bar{\vec{d}}_p, \qquad \hat{\mathbf{T}} = \Big[\mathrm{cov}_p(\bar{\vec{d}}_p) - \mathrm{mean}_p\, \mathbf{S}/n_p\Big]_+,$$

where $[\cdot]_+$ projects onto positive semidefinite matrices. That is a method-of-moments estimate: the spread of participant means minus the part their own sampling noise explains derived.

Participant offset. Given the data, $\vec{b}_p \sim \mathcal{N}(\hat{\vec{b}}_p, \mathbf{V}_p)$ with

$$\hat{\vec{b}}_p = \mathbf{T}(\mathbf{T} + \mathbf{S}/n_p)^{-1}(\bar{\vec{d}}_p - \vec{\mu}), \qquad \mathbf{V}_p = \mathbf{T} - \mathbf{T}(\mathbf{T} + \mathbf{S}/n_p)^{-1}\mathbf{T},$$

the standard Gaussian shrinkage: a participant with few samples is pulled toward the population average derived.

Bank. One bank of $N = 50$ pairs $(\vec{z}_n, u_n)$, $\vec{z}_n \sim \mathcal{N}(\vec{0}, \mathbf{I}_6)$, $u_n \sim \mathcal{U}(0,1)$, is drawn once with a fixed seed and **shared by every participant** read. With $\mathbf{C}_p = \mathbf{S} + \mathbf{V}_p$ (+ a small ridge), $\mathbf{L}_p\mathbf{L}_p^\top = \mathbf{C}_p$ and $\gamma_n = F^{-1}_{\chi^2_\nu}(u_n)/\nu$,

$$\vec{d}_{pn} = \hat{\vec{\mu}} + \hat{\vec{b}}_p + \mathbf{L}_p\vec{z}_n/\sqrt{\gamma_n}, \qquad \vec{x}_{pn} = \vec{x}_{\mathrm{ref}} + \mathbf{R}_{\mathrm{ref}}\,\vec{o}_{pn}, \qquad \mathbf{R}_{pn} = \mathbf{R}_{\mathrm{ref}}\exp\big([\vec{w}_{pn}]_\times\big).$$

Draw n is therefore the same quantile of the placement distribution for everyone (common random numbers) derived. A stochastic segment s uses bank pose $\varpi_s(n)$ in draw n, with ϖ_s a fixed permutation per segment, so segments are not perfectly correlated within a person, yet only 50 new field simulations per participant and target are needed read.

Two honest notes. The predictive scale $\mathbf{C}_p$ adds a Gaussian uncertainty to a t scale and then draws from a t; the sum of the two is not exactly t, so this is an approximation derived. And "the placement gets more consistent toward the end of a session", which Soma described, is not in the model; the fill is stationary within a session Soma read.

Held-out check. Each recorded sample is hidden in turn, $\hat{\vec{b}}_p$ refitted without it, and its PIT computed as

$$\mathrm{PIT}_{pi} = F_{F(6,\nu)}\big(m^2_{pi}/6\big), \qquad m^2_{pi} = (\vec{d}_{pi} - \hat{\vec{\mu}} - \hat{\vec{b}}_p^{\,(-i)})^\top \mathbf{C}_p^{-1}(\cdots),$$

since for a 6-dimensional t_ν, $m^2/6 \sim F(6,\nu)$ recalled. A calibrated model puts about 80% of PITs in $[0.1, 0.9]$ derived. The Gaussian version put only 41% there, with half of all PITs below 0.1, which is what heavy tails do to a Gaussian read; the t version reached 81% on synthetic t_3 data read. Its value on the real population is not reported in this draft. The bank size $N = 50$ is a choice Soma has questioned; it fixes the Monte Carlo resolution of every downstream quantity Soma inferred.

Scale. Within-participant SDs of (o_x, o_y, o_z) were 2.3, 1.7 and 4.3 mm and of (w_x, w_y, w_z) 2.1°, 2.3° and 5.0°; between-participant SDs were 0.5, 0.5, 2.2 mm and 0.9°, 1.0°, 1.9° (Gaussian first pass) read. So most placement variability is from session to session, not from person to person, and the largest is rotation about the coil's own normal derived.

9. E-field snapshots

For each recorded pose j of participant p, SimNIBS gives the field $\vec{E}_j(v) \in \mathbb{R}^3$ on central-surface vertices v of a **region of interest about 70 mm around the target**, at the coil current slope $\dot{I}_0$ used when the cache was built read recalled (Thielscher et al. 2015). These already existed for every participant before this work read.

For every bank pose the participant needs, one SimNIBS simulation with the same coil model and $\dot{I}_0$ is run on the same head model, and the field is sampled at the same ROI points, so bank and recorded fields are directly comparable read. Until a bank pose is simulated, it borrows the field of that participant's recorded pose nearest to it in Mahalanobis distance, labelled as a proxy, and is replaced automatically when the simulation lands read.

Two limits matter for everything below. **The ROI covers about 70 mm around the target**, so keys outside it have no field, and coverage is reported per key, not filled in read. **The volume.** The field is also stored on a 1 mm voxel box around the target for every pose, now for all participants including P115 and P116 read. Section 11 uses the surface and the volume together: the cortex is one layer, and the volume below it four more. The volume box has the same truncation as the surface ROI; at its edge the field is still about a fifth of its 99.5th percentile, so the edge shows in the figures and only full-head simulations would remove it read.

10. From pulses to contribution

This is where the two arms meet. The model is **one treatment simulation shared by all participants**: the same rules, the same bank, the same summaries; only anatomy, placement and run-sheet values differ Soma.

Key frame. The field lives on the head model it was simulated on, so the grid frame (centre, rotation, key assignment) must come from the same head model: from the population run for a population member, else from that brain's keyed-against-template record read.

Pulses add, fields do not. A pulse lasts well under a millisecond and pulses are about 110 ms apart, so a session's exposure is a sum over pulses, not a vector sum of fields derived. Each pulse contributes the magnitude of its pose's field, scaled by the stimulator output $O_k/100$. That factor uses the linearity of the field in the coil current slope and assumes the cached slope corresponds to 100% output; the mapping has not been read from the cache derived inferred.

Timing at IAF. Bursts of 40 pulses are timed at a frequency drawn from $\mathcal{N}(f_k, \epsilon_k^2)$ around the session IAF Soma read. The timing itself does not change the dose; the frequency enters only through the variants of section 12.

An earlier readout summarised the field per key and correlated each key's day dose with the day's clinical change. It was dropped: per key it was little more than the total day dose, and the question is about spatial covariance Soma read.

11. Contribution on the grid plane

The analysis target is the dose on the unwrapped grid itself, the plane (φ, ζ) of section 4, with the volume under the cortex included from the start Soma. The grid and the treatment timeline are, in this sense, a reformatting of the data: every pulse is annotated and keyed to a place on the plane, and anything else that can be put on the head model can be keyed on top of it later Soma.

11.1 Pose maps

The plane is cut into bins of 2° in φ by 0.02 in ζ, 180 × 100 over the sphere; because $\zeta = \cos\psi$, every bin has the same solid angle read derived. Each point $\vec{y}$ of the brain gets a direction, hence a bin b, and a **layer** ℓ by its depth $t = (\rho_2 - r)/(\rho_2 - \rho_3)$ along its ray: $\ell = 0$ is the cortex (central-surface vertices, area-weighted), and $\ell = 1, \ldots, 4$ are the volume at $t \in [0, \frac{1}{3})$, $[\frac{1}{3}, \frac{2}{3})$, $[\frac{2}{3}, 1)$ and $[1, \infty)$, the last being below the white-matter envelope read. For pose j of participant p,

$$\mathbf{m}_{p,j}(\ell, b) = \operatorname{mean}\left\{ \|\vec{E}_j(\vec{y})\| : \vec{y} \in \text{layer } \ell,\ \text{bin } b \right\}.$$

The **coverage** $c(\ell, b)$ is the fraction of pulses whose pose reaches the bin at all. Bins with partial coverage are drawn faded and are never filled read Soma.

11.2 Contribution

With $w_p[k, n, j]$ the number of pulses of session k delivered under pose j in draw n of the shared bank (section 8.4), and a session weight $\omega(k)$ (section 12),

$$X_{p,d,n}(\ell, b) = \sum_{k \in d} \omega(k) \sum_{j} w_p[k, n, j]\, \mathbf{m}_{p,j}(\ell, b), \qquad X_{p,d}(\ell, b) = \mathrm{median}_n\, X_{p,d,n}(\ell, b).$$

The **unit of dose is one pulse**: every one of the 3000 triggers of a session is carried in the data with its own time and pose, and the trains that appear in the figures are only a display grouping Soma read. Missing stretches of a session are filled from the empirical distribution of the history up to that point, through the placement model and the shared bank Soma. Units are V/m × pulses at the cached coil current slope, so only ratios mean anything until the output scaling of section 10 is confirmed inferred.

The full vector. Cortex and the four volume layers together are one vector per participant-day, and the cortex is also analysed on its own Soma. Both are written back onto the anatomy: a per-vertex map (bin, key, layer) puts the plane on the cortical surface as a surface file, and every grey- and white-matter voxel of the head model's tissue volume is given its key, bin and layer, so the contribution exists as a volume image as well read. The same maps run the other way, so any other spatial data on the head model, for example another imaging modality, can be binned onto the plane and enter the models as a covariate or an outcome derived.

11.3 What the plane shows

Figure 5 is the contribution of P116 on the plane. Two things are visible at once. The contribution is concentrated in a few keys around the target and spreads further in the outer volume layer than on the cortex, and the five days look almost the same: for this participant every day is within 3 to 8% of the mean day read. Day-to-day change in the dose is small next to its spatial structure, so the spatial share and the total have to be separated before anything is compared with a clinical change derived.

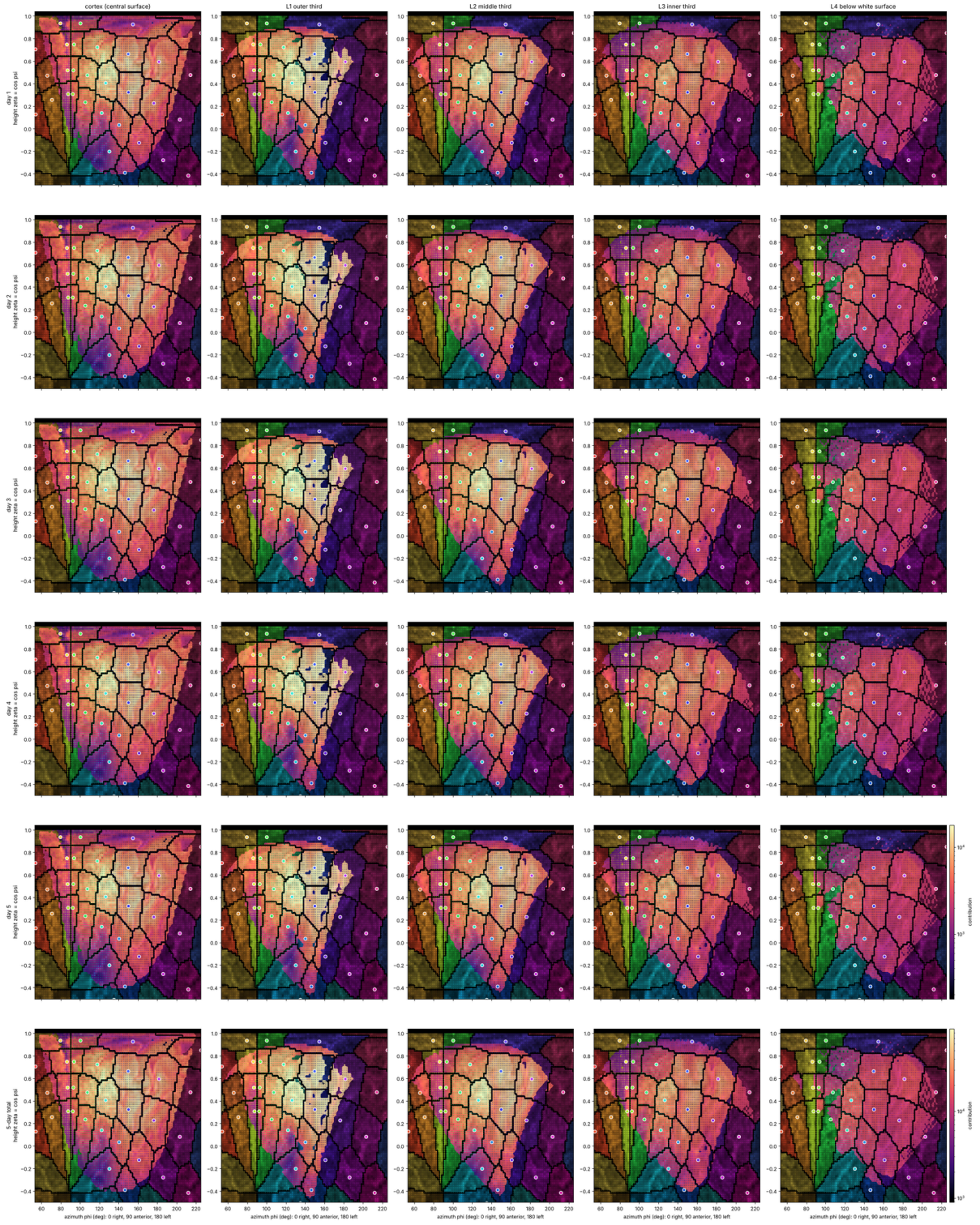


Figure 5. Contribution of P116 on the grid plane. Rows: treatment days 1 to 5 and the 5-day total. Columns: cortex and volume layers L1 to L4. Colour: contribution (output-scaled, median over the pose draws) on a log scale over two decades, one scale for all layers and days, the total on its own; faded = only part of the pulses reach the bin. Underneath, the key colours, shaded by the gradient of the tissue integral I so that the folding stays visible. White dots: key centres. Azimuth 0° right, 90° anterior, 180° left.

Other views of the same data read. Two further figure sets are made per participant and are not reproduced here. *Depth structure*: per bin, the correlation over the coil poses (pulse-weighted) between the field in one layer and in the layer below, with its slope, and the correlation of a bin with its neighbours in the same layer. Under the target the field passes downward almost unchanged (r near 1), and neighbour correlation is near 1 almost everywhere because the field is smooth over poses at this resolution, so it says little inferred. *Real space*: the contribution returned to sagittal and coronal slices of the head model, 15 mm apart through the contribution centroid, viewed from the coil side, coloured by

direction from C and brightened by contribution ($0.12 + 0.88\,(c/c_{99.5})^{0.75}$, no hard cut-off) Soma. Slices are cut in AC-PC axes, taken as the rotation of a least-squares affine fit to the head model's own warp to MNI space; the fits have 1.9 to 2.8 mm residual and remove a pitch of 7.6° to 24.2° that the earlier scanner-axis slices carried read. Per-day versions of the participant figures are no longer made: the days differ too little to need them, and the figures now serve the covariate maps Soma.

12. Control models and variants

12.1 What a variant is for

The treatment is fixed: the same pulses, the same poses, the same fields. A **variant** changes the predictor in a way that keeps the treatment timeline a reasonable approximation of what happened, and the question is **whether the relationship between the predictor and the clinical and comfort covariates changes when it does**. It is not whether some variant explains the outcomes better Soma. Variants run alongside the main path; they never replace it Soma.

Soma frames this as an **empirical Euler–Lagrange** problem Soma. Given an input scheme S (the full run, a permutation, or a leave-one-out), the model fitted to the actual path fixes one trajectory: the slope of each clinical measure over the treatment days. A variant is a deviation of the path, and the readout is how the fitted parameters move. Written out inferred: with $\hat{\theta}_S(X) = \arg\min_\theta \mathcal{L}(\theta;\, X, Y, S)$ the fit on the actual contribution X, a variant $X^v = X + \delta X$ is read through

$$\Delta\theta_S^v = \hat{\theta}_S(X + \delta X) - \hat{\theta}_S(X),$$

and through the change in the predicted correlation of the trend with the clinical measures and with comfort. Entropy measures and the covariates are what validate the dynamics Soma. "Euler–Lagrange" is used here as an analogy for comparing paths around the actual one; no action functional is defined, and the formal content is the parameter shift above inferred.

Because the fields and the pulse weights w_p are stored separately (section 11.2), a variant that only changes ω or the pose draws costs no new field simulation derived.

12.2 Session-weight variants

With O_k the stimulator output (%), f_k the IAF of session k, ϵ_k the run-sheet RMSE read in Hz as in Soma's model $f \sim \mathcal{N}(f_k, \epsilon_k^2)$, $\bar{f}_p$ the participant's treatment-mean IAF, and a tuning width σ read:

Variant	$\omega(k)$	Meaning
m0 pulses	1	the pulses alone, as any protocol delivering them would; no IAF
m1 output	$O_k/100$	the field scales linearly with output
m2 sync	$\dfrac{O_k}{100}\,\dfrac{\sigma}{\sqrt{\sigma^2+\epsilon_k^2}}$	gain of a Gaussian tuning of width σ around the session IAF, averaged over $f \sim \mathcal{N}(f_k, \epsilon_k^2)$
m3 stable	$\mathrm{m2} \cdot \exp\!\Big(-\dfrac{(f_k-\bar{f}_p)^2}{2(\sigma^2+\epsilon_k^2)}\Big)$	the same tuning centred on the participant's mean IAF; sessions whose IAF drifted count less

The m2 gain is exact: $\mathbb{E}_{f\sim\mathcal{N}(f_k,\epsilon^2)}\left[e^{-(f-f_k)^2/2\sigma^2}\right] = \sigma/\sqrt{\sigma^2+\epsilon^2}$, a Gaussian convolution, and centring on $\bar{f}_p$ multiplies it by the factor in m3 derived. $\sigma \in \{0.5, 1, 2\}$ Hz is chosen per outcome and counted as a parameter read. If the run-sheet RMSE is a phase error rather than a frequency spread, ϵ_k needs a conversion before m2 and m3 mean what they say inferred.

IAF-recalculated variants. The IAF used in treatment came from the closed-loop system's own estimate. The same rest recording before each session is re-analysed offline to give two further IAFs, one from occipitotemporal and one from frontal channels, each with its spread over the recording Soma read. These enter as a second axis, the **IAF source** $\in$ {treatment, occipitotemporal, frontal}, replacing (f_k, ϵ_k) in m2 and m3; m0 and m1 do not depend on IAF and are unchanged read derived. Everything else stays: the same trains, poses, draws and fields. The variants therefore live in **the same sample space** as the actual path and get exactly the same analytic treatment, which is what makes a difference between them attributable to the IAF alone Soma derived. One limit follows from the same property: since all IAF sources share the pose draws, pose uncertainty cancels in the comparison between them but still sits inside each correlation derived. Offline, each region's IAF is estimated from the ICA-cleaned rest recording by a Whittle likelihood on its periodogram, with an aperiodic power law plus a Gaussian alpha peak, and how finely the channels are partitioned into regions is chosen by AICc read. These values are not yet joined to the variants.

12.3 Deviation classes

Soma's deviations fall into two classes Soma.

Class 1: no protocol deviation, only more variance. The protocol is kept; the uncertainty about it is widened.

- *Wider pose spread.* More trains are treated as stochastic, or the candidate spread of the placement model is inflated, for example $\mathbf{C}_p \to \lambda\,\mathbf{C}_p$ with $\lambda > 1$ Soma inferred. This is also the natural answer to the sampling problem of section 8.2: rather than resolving where the coil was in 2220 gaps from about 28 samples, the fill is treated as a variant, and the check is whether the dose spread across draws is small next to the spread between participants inferred.
- *IAF shift as temporal scrambling.* The point of shifting IAF is to test whether **scrambling the dose in time across the days** makes a significant difference against the correlation on the actual path Soma. With π a permutation of a participant's sessions, the scrambled variant reassigns the session weights, $\omega^{\pi}(k) = \omega(\pi(k))$, which keeps the 5-day total and moves dose between days inferred. Both the actual and the variant contribution are correlated with the outcomes, and with each other, so the comparison is encoded numerically by the structure of the treatment itself: how far the variant moved the predictor ($\mathrm{corr}(X, X^{\pi})$) is known alongside how far the relationship moved Soma. **Permutations only reshift the variant weights**; the pulses, poses and fields are never permuted Soma.

Class 2: move the target. The contribution is moved to a nearby region on the sphere, $X^{R}(\ell, \hat{u}) = X(\ell, \mathbf{R}^{-1}\hat{u})$ for a small rotation $\mathbf{R}$ inferred. Under the unbiased grid, the trends should then be the same across participants up to variance Soma. This is a check of the grid rather than a hypothesis about treatment; it is the clinical counterpart of the rotation null in section 5.1 derived.

12.4 Variants along the population modes

The variants above are the same rule for everyone. Soma's intended family is **individual** Soma: the contribution distribution is modelled in several ways, including the empirical distribution itself as an input, and the variants are combinations along the dominant modes of the population (the population mean and its distribution, which Soma calls the "curl"). Each participant is then a grid of individualised deviations from that structure, and a variant reallocates contribution toward an alternative protocol or a nearby region along those modes Soma.

One way to write this down inferred: with $s_{p,d} = X_{p,d} / \sum X_{p,d}$ the spatial share, $\bar{s}$ its population mean and $\mathbf{U}$ the leading principal directions of the shares, participant p has coordinates $\mathbf{a}_{p,d} = \mathbf{U}^{\top}(s_{p,d} - \bar{s})$ and residual $r_{p,d} = s_{p,d} - \bar{s} - \mathbf{U}\mathbf{a}_{p,d}$. A variant moves the coordinates, $s^{v}_{p,d} = \bar{s} + \mathbf{U}(\mathbf{a}_{p,d} + \boldsymbol{\delta}_p) + r_{p,d}$, and the permutation null reshuffles only $\boldsymbol{\delta}$. Whether "curl" means these modes or a rotational part of a flow on the plane has not been confirmed, and this family is not built yet.

12.5 Model, information criterion, first result

Rows are participant × day with a clinical change. Each variant's contribution is reduced to the **same low-dimensional basis**, fixed before any outcome is looked at, and the variants are compared on **identical rows** Soma read:

$$\Delta Y_{p,d,o} = \alpha_p + \boldsymbol{\beta}^{\top}\mathbf{Z}\left(s^{m}_{p,d}\right) + c\,\log T^{m}_{p,d} + \gamma\,\mathrm{comfort}_{p,d} + \varepsilon,$$

with $T^{m}_{p,d}$ the total, α_p participant intercepts, $\mathbf{Z}$ the first three principal components of the m0 shares over key × layer features (the same loadings for every variant), and comfort the day mean of the standardised momentary comfort items read. Variants are ranked by

$$\mathrm{AICc} = -2\log\hat{L} + 2K + \frac{2K(K+1)}{n-K-1}, \qquad w_m = \frac{e^{-\Delta_m/2}}{\sum_{m'} e^{-\Delta_{m'}/2}},$$

and the spatial term is tested on its own as AICc(full) − AICc(without $\mathbf{Z}$), against 200 permutations of ΔY within each participant read recalled (Burnham and Anderson 2002). Keeping the basis and the rows fixed is what makes this a comparison of variants and not of feature sets derived.

First population run read: 60 participant-days, of which 48 rows from 12 participants have a change; 5 outcomes; 17 comfort items; the three components carry 58%, 29% and 4% of the variance of 68 key × layer shares. m0 and m1 are tied within 1.7 AICc units on every outcome but one, where m0 is better by 4.9. m2 and m3 cost 5 to 7 units, which is the price of σ with no gain in likelihood. The spatial components do not earn their parameters on any outcome (ΔAICc +5 to +14); the smallest permutation p is 0.07. With $n \approx 48$ this says the spatial term is not supported **yet**, not that it is absent derived. This run used the treatment IAF only. The covariate analysis of section 13 replaces the per-outcome figures that came with it.

13. Export and covariate analysis

13.1 Tables

Every participant with a finished grid gets two tables, rebuilt whenever a participant is added read:

- **Contribution**: one row per day × variant × key × layer, with the contribution summed over the day's six sessions; its share of the day total; the share relative to the population (ratio, and z over all participant-days for the same variant, key and layer); the key centre's angles; the day's pulses, output, IAF and RMSE; the pose mix (number of poses, effective number e^{H} with H the entropy of the pose weights, top pose and its share); comfort; and the clinical day values (score, next-day score, change).
- **Pose**: one row per coil pose with its orientation and its weight per day. Orientation comes from Soma's existing neuronavigation extraction, which already gives turn, tilt and twist, their deviations from target and the target's own values, and the arc, depth and angle errors; it is joined to the simulated poses by session and sample rank Soma read. Day values are pose-weighted means, circular for turn and twist read.

Population tables hold the mean, SD and count of each share per variant, key and layer, the model rows and the information-criterion table read. A per-variant sample table, one row per participant-day with the covariates and the log-ratio share of every valid cell, is written for the classifier work read.

13.2 Covariate maps

For each covariate x (each clinical change, comfort, output, IAF, RMSE, pulses, the effective number of poses) and each grid cell c (key × layer), the map shows how the cell's share moves with x **within participants** read. The first version demeaned share and covariate per participant only and drew cells at uncorrected $p < 0.05$ over 1694 tests. That has two faults: anything that drifts with treatment day correlates with a change score regardless of mechanism, and about 85 cells pass by chance inferred. The current version read:

1. **Log-ratio shares.** Shares sum to one, so they are negatively coupled by construction; each is replaced by its centred log-ratio, $\mathrm{clr}(s)_c = \log s_c - \frac{1}{C}\sum_{c'} \log s_{c'}$ recalled (Aitchison 1986).
2. **Two-way fixed effects.** Participant and day effects are removed from both sides before correlating, so r is within-person and detrended over days, with $n - P - D$ degrees of freedom.
3. **Family-wise error.** Each covariate is permuted within participant 1000 times and the maximum $|r|$ over all cells is kept; a cell is drawn opaque only if its r exceeds the 95th percentile of that maximum.
4. **ICC per cell.** The share of variance between participants against within, which says whether a cell can follow days at all or only separates people.
5. **Nested baselines.** On the same rows, by AICc: participant effects alone; + total dose; + the top key; + the three grid components; + total and components. This is the test of whether the spatial part adds anything beyond pulses × output.

The maps are drawn on the reference brain in the real-space layout of section 11.3. **Variants** are compared on the same footing: for each variant the same statistics are computed, and the difference $r_{\text{variant}} - r_{\text{m1}}$ per cell gets a participant-bootstrap 95% interval and a Benjamini–Hochberg q read. This is the numerical form of the readout of section 12.1: how far a variant moves the relationship, against how far it moved the predictor.

State of the evidence. The first covariate maps were made with the first version and are exploratory only inferred. With about 60 participant-days few or no cells are expected to survive the family-wise test inferred. The v4 statistics and the per-variant comparison are running at the time of writing and have no result in this draft.

14. How it runs, and the state on 9 November

Execution. Everything runs on one analysis machine (23 cores, about 110 GB usable memory) under Soma's batch controller read. Each stage is a driver that declares one job per participant with its inputs, a memory estimate, a thread count and a done marker. The controller admits a job only when its inputs exist and it fits; a job whose inputs are missing waits outside the CPU budget; a finished job is never rerun read. Each stage runs as its own controller instance with its own memory cap, because a job cannot be added across instances read. Phase 1 and phase 2 run as two separate sets of instances, so that the time each stage takes can be compared between them Soma.

Order per participant Soma read: recordings converted → timeline, samples and pose decisions → bank-pose fields → dose on LSL timing → grid contribution and variants → volume images → export, covariate maps and upload → recordings published to network storage. A new participant enters by appearing in the raw recordings folder; every later stage picks it up when the previous stage's done marker appears. The figure and export stage now waits for each new grid and runs the export, the covariate and validation maps and the upload without a manual step read.

Where results go. One shared folder holds figures (per participant, population, covariates) and data (per participant, population, per variant); everything from before 9 November was moved, not deleted, into a superseded folder read. Paths, instances and done markers are listed in the companion path reference.

State on 9 November, about 04:45 UTC read:

Stage	Phase 2 (12)	Phase 1 (20)
Head models, keyed	done	done (archive batch)
Recordings converted	done (one day being redone)	running, staged 3 at a time (about 100 GB raw each)
Pose decisions, bank fields, dose	done	14 queued behind conversion; 5 need neuronavigation extraction and field caches
Grid contribution, variants m0–m3	done	waiting
Volume images, real-space figures	done; AC-PC re-render running	waiting
Population model (AICc)	first run done	—
Export, covariate maps (v4 statistics)	running	—
IAF-source variants	waiting for the offline IAFs	—
Published to network storage	running	waiting

15. What is not solved

About the model.

- **Dose is linear and additive**, with $||\vec{E}||$ standing in for effect: no saturation, no plasticity, no dependence on field direction relative to the cortex. These are modelling choices, stated, not errors inferred.
- **Little within-person variation.** Days differ by a few percent in a participant's contribution, and the first population run found no spatial term worth its parameters. A quantity that barely changes from day to day cannot follow day-to-day clinical change; the per-cell ICC will say how much of the grid separates people rather than days derived.
- **Keys are not person-discriminative** (section 5), so no claim should rest on key identity alone.
- **Uncertainty is collapsed.** The 50 draws are reduced to a median before the statistics, and the field truncation at the ROI edge leaves an edge at about a fifth of the 99.5th percentile.

About the inputs.

- **Field coverage.** About 70 mm around the target on the surface and a box in the volume; full-head simulations would be needed for any claim about keys far from the target.
- **Output scaling.** The mapping from the cached coil current slope to % output is assumed, not read.
- **RMSE units.** Whether the run-sheet RMSE is a frequency spread or a phase error is not settled; m2 and m3 depend on it.
- **The 0.1 s gaps** (at least a tenth of all gaps) are back-to-back or split trains; not checked.
- **Brainsight clock residual** δ is not identified; it can move samples across the 5-minute window.
- **Target.** Deviations are measured against the target stored in Brainsight; that this equals the pipeline's dlPFC point is assumed.
- **Within-session tightening** of placement is not modelled.

About the grid. A leave-one-out template test, a centre-translation test (a 3 mm shift turns boundaries by about 2.6°) and MRI test–retest are still missing. Distance from the centre is not cortical depth, so laminar work needs an equivolume depth recalled (Waehnert et al. 2014). The folding index mixes folding with ray obliquity.

Open specifications. The population-mode variants need Soma's definition of the modes; the ideal-transfer reference coil pose is proposed and waiting for a decision.

16. What changed from the earlier specifications

Earlier	Now	Why	Decided by
centre at the commissure midpoint	centroid of the pial-enclosed volume	the registration misplaced it by up to 4 cm	Claude, accepted by Soma
grid refined at the target	uniform grid	a prefrontal bias in the predictor	Soma
seeds from each person's cap	one template in each person's anatomical frame	cap placement moved keys by about 10°	Soma (pole), Claude (template)
pause = gap from file-name times	restarts between runs + LSL gaps > 60 s	file-name latency faked continuous joins	Soma
per-participant Gaussian fill	population t placement model, shared bank	unbiased, comparable across people	Soma
field at the target per day	dose per key per day	the question is the spatial encoding	Soma
dose per key	dose on the grid plane, cortex + four volume layers	spatial covariance, not key totals	Soma
one weighting	a family of variants on the same path, compared by AICc	control models: same treatment, different bias	Soma
run-sheet IAF only	IAF source as a variant axis (treatment, occipitotemporal, frontal)	same sample space, IAF effect isolated	Soma
covariate r demeaned per participant, uncorrected p	log-ratio shares, participant and day effects removed, family-wise permutation, ICC, nested baselines	day-trend confound and 1694 tests	Claude, accepted by Soma
slices in scanner axes	AC-PC axes from the head model's MNI warp	sagittal cuts were tilted by up to 24°	Soma
per-day participant figures	per-participant figures feed the covariate maps; days not re-rendered	days differ by a few percent	Soma

Appendix A. Scripts

Every stage runs a script on the analysis machine under its own controller instance; the companion **path reference** lists, per stage, the instance, inputs, outputs and done marker. md5 is the first 8 hex digits of the copy in the project folder on 9 November at 04:50 UTC. Where the machine holds a later patched version, it is noted.

Stage	Driver	Worker scripts	md5 (project copy)
shell grid	—	`shellpop.py` (population, `--reference` for new brains)	672d517a
participant set (phase 1 / 2)	`pset.py` (imported by the drivers)	—	bc8a0fab
recordings	`xdf30.py` (phase 2 per day; phase 1 inside the pose driver)	—	4bd3be44 (machine: patched, skips empty stub files)
timeline + pose decisions	`rt_pro.py`	`tl_xdf.py`, `bs_key.py`, `bs_gaps.py`, `bs_pauses.py` v7.2	7d46b843; b3464934, 3821b9af, d82f6c3a, c20a5068
bank-pose fields	`bank_pro.py` v2	`bank_sim.py` v2	94545e91; 067ca8bd
dose on LSL timing	`dose_pro.py` v3 (`dose1_pro.py` first pass)	`keydose.py`	bb8733f0 (11a6ed0e); b02900a7
grid contribution, variants, population model	`grid_pro.py`	`gdose.py` v4, `gvar.py`, `gpop.py`	2fc52cc9; f36e7bcb, 2d149575, d37dac89
volume images	`gembed_pro.py`	`gembed.py`	709135a7; 811f4cf3
export, covariate maps, real-space figures, upload	`gvol_pro.py` (`gcontrib_pro.py` for the first export)	`gexport.py`, `gvol.py`	542331d2 (1aee8160); e7a9f9f9, 633bd0b8 (machine: AC-PC and v4/v5 statistics patches)
publication	`publish_pro.py`	`xdf30.py --annotate`	c6337d87
offline IAF	(phase-1 chain)	`iafwhittle.py`, `iafmerge.py`	7a27eb53, 6eda106d

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